Continuous Compounding & Euler's Constant e

Exponential & Logarithmic Functions • Standard: CCSS.MATH.HSF.LE.A.4

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate discrete and continuous exponential growth models from the inside out (exponent first, then base power, then initial value).

Exponential growth $P \cdot e^{rt}$ compounds continuously by scaling a base quantity according to an inner rate-time product ($r \times t$). Evaluating from the inside out—computing the exponent $r \times t$ first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (Logarithmic Blocks) & Growth Curve

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).

Student Prediction (Before Calculator): Invest $1,000 at 7% continuous interest for 10 years: $A = 1000 \times e^{0.07 \times 10} = 1000 \times e^{0.7}$. Will it double? ($e^{0.7} \approx 2.01$).

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: `0` `.` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
`0` `.` `0` `7`	`0.07`	Rate r = 0.07 entered in X
`ENTER`	`0.07`	Rate lifted to Y
`1` `0`	`10`	Time t = 10 entered in X
`×`	`0.7`	Exponent $r \times t = 0.70$
`e^x`	`2.0138`	$e^{0.70} = 2.01375$ (Growth factor: more than doubled!)
`1` `0` `0` `0`	`1000`	Principal P = 1000 (growth factor lifts to Y)
`×`	`2013.75`	Total accumulated amount A = $2,013.75!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (John Napier's Logarithms: Turning Multiplication into Height Addition)

In 1614, Scottish mathematician John Napier realized that logarithms turn tedious multi-digit multiplication into simple addition—the exact engineering principle behind our Prime Factor Studio blocks, whose physical heights scale as $h = 15 \times \log_2(p)$ mm so stacking blocks adds their logarithms! Paired with Jakob Bernoulli's continuous compounding constant e, logarithms unlock doubling time and half-life.

Core Investigation (Student Warm-Up A):

A bacteria culture starts with P = 500 cells and triples every hour ($A = 500 \times 3^t$). Calculate the population after t = 4 hours.

Solving for Time: Doubling Time, Half-Life & Natural Logarithms

Logarithms & Inverting Exponentials • Standard: CCSS.MATH.HSF.BF.B.5

1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

Concept & Stack Approach: A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height ($\ln(a \cdot b) = \ln(a) + \ln(b)$). To solve for time t in $P \cdot e^{rt} = A$, divide A by P first, take the natural logarithm $\ln(A/P)$ to unwrap the exponent, and divide by r.

Blocks & Prediction: Observe how Prime Factor block heights scale with ln(p) on the comparator lid while comparing additive block heights to multiplicative products. Prediction: At 5% continuous growth (r = 0.05), doubling time is t = ln(2) / 0.05 ≈ 0.69315 / 0.05 ≈ 13.86 years.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[2]	**2**	Doubling target factor 2
[LN]	**0.6931**	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	**0.05**	Rate r = 0.05 (ln(2) lifts to Y)
[÷]	**13.86**	t = ln(2) / r = 13.86 years!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as $2 \times 5 = 10$) or reversing Base and Exponent on [y^x].

Hands-On Block Remediation: Stack five [2] Prime Factor logarithmic blocks in the comparator lid to show that adding 5 block heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in $P(1 + r)^t$ (computing $(P \cdot (1+r))^t$ instead of $P \cdot ((1+r)^t)$).

Hands-On Block Remediation: Use the Inside-Out RPN rule: compute $(1 + r)^t$ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

Extension Challenge (Student Warm-Up B):

A radioactive isotope decays according to $N(t) = N_0 \cdot e^{-0.035t}$. Find its half-life t = ln(2) / 0.035 to two decimal places.

Expected Answer: t = ln(2) / 0.035 ≈ 0.6931 / 0.035 ≈ 19.80 years.

Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Discrete exponential growth multiplies by a constant factor each period: $A = P \times b^t$ (or $A = P \times (1 + r/n)^{n \cdot t}$).
- Continuous exponential growth follows $A = P \times e^{r \cdot t}$, where Euler's constant $e \approx 2.71828$ is the base of natural growth.
- Logarithms invert exponentials ($\ln(e^x) = x$) and turn multiplication into addition: $\ln(a \times b) = \ln(a) + \ln(b)$.

Key Sequence: `0` `.` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Investigation 1: Evaluate e^2 to 4 decimal places.

1. Growth / Decay Timeline Sketch:

2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Find the continuous doubling time $t = \ln(2) / r$ for an investment growing at $r = 8\%$ (0.08).

1. Growth / Decay Timeline Sketch:

2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Evaluate $10^{3.5}$ (note that $10^3 < 10^{3.5} < 10^4$).

1. Growth / Decay Timeline Sketch:

2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Solve $10^x = 500$ for x using the base-10 logarithm log10(500).

1. Growth / Decay Timeline Sketch:

2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 10: Exponential Growth, Compound Interest & Logarithms — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 10–12

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A bacteria culture starts with P = 500 cells and triples every hour ($A = 500 \times 3^t$). Calculate the population after t = 4 hours.

Core Solution: $3^4 = 81$; $500 \times 81 = 40{,}500$ bacteria.

Core Check Keys: [3] [ENTER] [4] [y^x] [5] [0] [0] [×]

Extension Challenge (Warm-Up B): A radioactive isotope decays according to $N(t) = N_0 \cdot e^{-0.035t}$. Find its half-life $t = \ln(2) / 0.035$ to two decimal places.

Extension Solution: $t = \ln(2) / 0.035 \approx 0.6931 / 0.035 \approx 19.80$ years.

Ext. Check Keys: [2] [LN] [0] [•] [0] [3] [5] [÷]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Evaluate e^2 to 4 decimal places.

Expected Answer: 7.3891

SC-32 Verification Keys: [2] [e^x]

Investigation 2: Find the continuous doubling time $t = \ln(2) / r$ for an investment growing at r = 8% (0.08).

Expected Answer: 8.66 years

SC-32 Verification Keys: [2] [LN] [0] [•] [0] [8] [÷]

Investigation 3: Evaluate $10^{3.5}$ (note that $10^3 < 10^{3.5} < 10^4$).

Expected Answer: 3162.28

SC-32 Verification Keys: [3] [•] [5] [↶] [10^x]

Investigation 4: Solve $10^x = 500$ for x using the base-10 logarithm log10(500).

Expected Answer: 2.6990

SC-32 Verification Keys: [5] [0] [0] [↶] [LOG]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.